

Mario Klobučarić

AI Engineer & Consultant

Profile

AI Engineer building production LLM applications, RAG systems, and agentic workflows. Core expertise in Python, FastAPI, LangChain, LangGraph, and cloud deployment (GCP, AWS).

15+ years of prior experience in project management, business development, and technical consulting – including €7.5M EU-funded projects. This background provides a comprehensive perspective on delivering AI solutions that solve real business problems. Recent work includes enterprise AI consulting for pharmaceutical R&D and national banking institutions, plus end-to-end development of complex AI-powered applications.

Professional Experience

AI Engineer & Consultant | techXflow d.o.o.

November 2023 – Present | Self-employed, Croatia

Building production AI systems and providing strategic AI consulting for enterprise clients.

Selected Projects (2024–2026):

SmartWay Design Portal (2025–2026)

Full-stack e-content platform for electrical engineering design company. Content store with bank transfer payment flow, newsletter system with block editor serving two brands, GDPR-compliant admin dashboard. React 18/TypeScript, Firebase (Auth, Firestore, Cloud Functions), Tailwind CSS.

Construction Coordination Platform (Q4 2025)

Full-stack app replacing manual Excel workflow for county government. FastAPI, React 19/TypeScript, PostgreSQL, Firebase Auth, GCP Cloud Run. Reduced search time from 10-15 min to <30 sec. 5-week delivery using Claude Code.

Banking Sector AI Consulting – HPB d.d. (2025)

Strategic AI education for board member and 15+ sector directors at Croatia's national-level bank. Covered practical AI implementation strategies and use cases.

Pharmaceutical AI Consulting & RAG PoC – Belupo d.d. (2024–2025)

AI strategy consulting and education for R&D department and directors (2025). Earlier RAG chatbot proof-of-concept for 160+ medication PDFs using LangGraph, Weaviate hybrid search, multi-LLM support (2024).

Medical Transcription System (2025)

Desktop app for Croatian medical dictations across specializations. Gemini AI integration, real-time transcription with auto-typing, template system. GCP deployment.

Energy Efficiency RAG Platform – VIREAS (2024–2025)

Agentic RAG system providing renewables and energy efficiency guidance for citizens, developed for regional energy agency. FastAPI + LangGraph, Weaviate hybrid search, Voyage AI reranking, multi-LLM support. Includes Prefect ETL pipeline and MCP server for Claude Desktop integration.

BOM Extraction System (2025)

Automated Bill of Materials extraction from German-language manufacturing documents. Landing.ai document extraction, multi-format output (Excel/PDF/JSON). Reduced 3-4 hour manual process to minutes.

Automated Content Generation (2025)

Blog automation with LangGraph 8-node workflow + Perplexity research integration. Custom Vertex AI fine-tuning pipeline for Gemini. Video pipeline for TikTok/Reels using N8N.

Text-to-SQL for Brewery Production (2025)

Natural language to SQL using LangGraph and open-source LLMs (Ollama). Query classification, SQL safety validation, auto-visualization with Plotly.

Contact

Remote/Hybrid
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Links

[LinkedIn](#)
[GitHub](#)
[Personal Page](#)

Skills

AI & LLM

Claude, GPT, Gemini
LangChain, LangGraph
CrewAI, LlamaIndex
RAG, Vector DBs
Prompt Engineering

Backend & Cloud

Python, FastAPI, Flask
PostgreSQL, SQL
GCP, AWS, Firebase
Docker, REST APIs
OAuth/JWT, N8N

Frontend

TypeScript, React
Flutter

Data Engineering

ETL Pipelines
OCR + LLM
Text-to-SQL

Languages

Croatian – Native
English – Professional
German – Intermediate

Certifications

Agentic AI (2025)
RAG – DeepLearning.AI
Gen AI Leader – GCP
Data Eng. AWS (2024)

StudyBuddy Educational App (2023)

Full-stack AI study assistant: Flutter cross-platform app, Flask backend, ChromaDB RAG, Firebase. Photo → OCR → AI-generated Q&A. Multi-language support.

Teacher & Head of Regional Competence Center

Technical School Čakovec
September 2018 – November 2023

Coordinated implementation of two EU-funded projects valued at €7.5 million (staff coordination, didactic equipment, procurements). Taught coding (Python, embedded C+ +), automation, robotics, and embedded systems.

Agency Director | Međimurska energetska agencija d.o.o.

January 2013 – January 2016
Directed agency operations, achieved complete financial self-sustainability. Managed multiple EU energy efficiency and renewable energy projects.

Earlier Experience (2006–2018)

Head of International Projects & Sales | ENERGY PLUS d.o.o. (2016–2018)
Licensed Electrical Designer | DORS PROJEKT d.o.o. (2010–2012)
Electrical Designer | ZG-PROJEKT d.o.o., Zagreb (2007–2010)
Development Engineer | V-elin d.o.o. (2006–2007)

Education

Graduate Electrical Engineer (Industrial Electronics)

Faculty of Electrical Engineering and Computing, Zagreb | 2000–2005

University Specialist of Strategic Entrepreneurship

Faculty of Economics, Zagreb | 2008–2011

Teachers Education Specialization

Faculty of Organization and Informatics, Varaždin | 2017–2018

Certifications

DeepLearning.AI: Agentic AI | October 2025
DeepLearning.AI: Retrieval Augmented Generation (RAG) | Coursera, 2025
Generative AI Leader | Google Cloud Certification, 2025
DeepLearning.AI: Data Engineering (AWS) | Professional Specialization, 2024